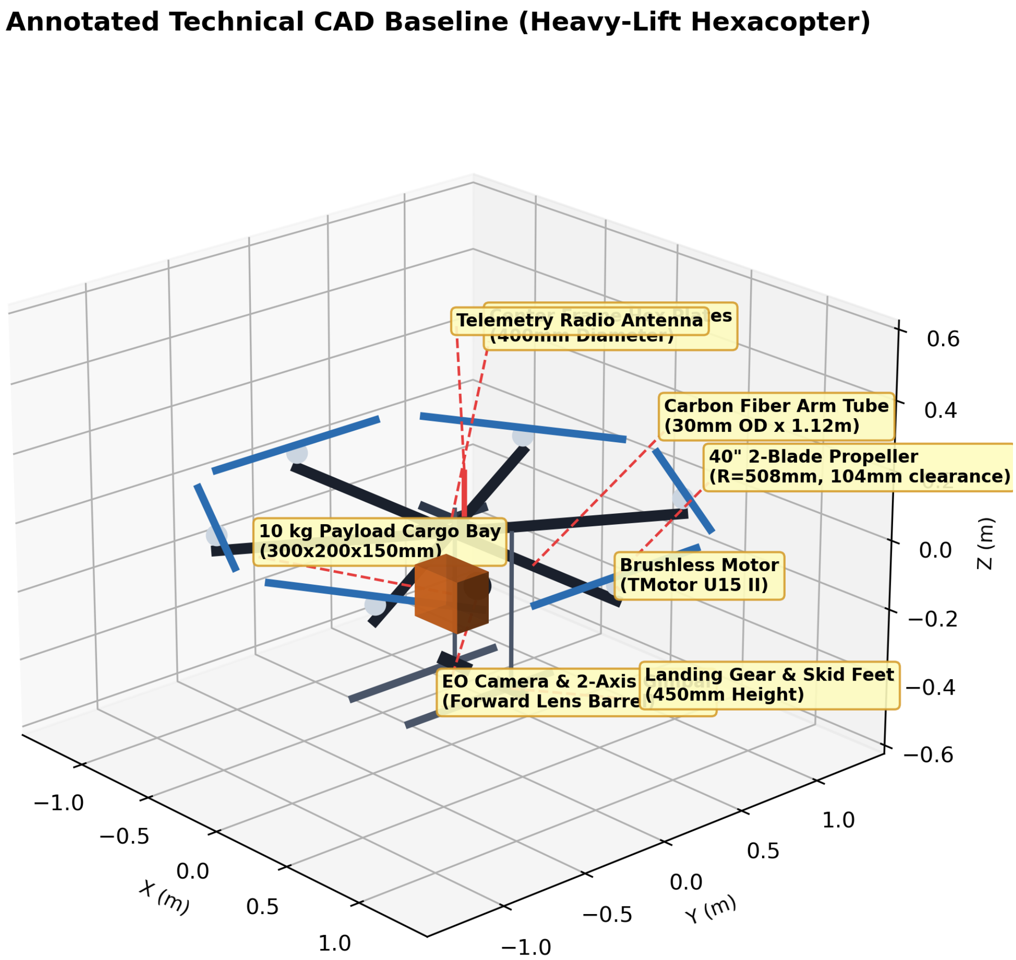
Engineering Design, Aerodynamic BEM, Structural FEA, and Mission Validation of a Heavy-Lift Multirotor UAV

Full Technical Baseline, FreeCAD/STEP 3D Assembly, XFLR5 Aerodynamics, and KiCad Electrical Schematic



Prepared by:
Flight Systems Engineering Team
IIT Madras — BS Electronic Systems

Document Control & Source Links:
Locked Design Baseline: `DESIGN_LOCK.md`
GitHub Repository: `projects/heavy-lift-uav`

Date: July 2026

Contents

1 Executive Summary & Mission Requirements

This report documents the engineering design and multi-physics validation of a heavy-lift multirotor Unmanned Aerial Vehicle (UAV). The vehicle is engineered to carry a **10.0 kg payload** over a **30 km operational mission radius** with a **20-minute hover loiter** and a **20% fuel reserve margin**.

The physical system consists of a **Single-Motor Hexacopter Topology** (6 arms at 60° radial spacing, 1 motor + 1 ESC + 1 40-inch propeller per arm) powered by a **Gas-Electric Hybrid Generator** (3.6 kW continuous output driving a 48V DC bus).

Table 1: Locked System Baseline Specifications (`DESIGN_LOCK.md`)

Parameter	Locked Value	Engineering Basis / Status
Aircraft Configuration	Standard Hexacopter	6 arms @ 60° spacing $(0°, 60°, 120°, 180°, 240°, 300$
Arm Tube Length (L)	**1.120 m** (1120 mm)	30mm OD × 2mm wall carbon tube (104 mm prop
Propeller Specification	**40” × 13”** Carbon Fiber	$R = 0.508$ m, 2-blade tapered profile, 8.1 g/W hov
Total Takeoff Weight (TOW)	**34.575 kg**	Converged via 5-iteration mass-power-fuel loop
Hover Electrical Power	**3,460.1 W**	Total DC electrical draw (576.7 W per motor bran
Carried Fuel Mass	**2.323 kg**	1.859 kg mission fuel burn + 0.464 kg (20% reserv
Governing Load Case	Asymmetric Motor-Out	1.5G Emergency Recovery ($\sigma_{\max} = 164.4$ MPa, SF

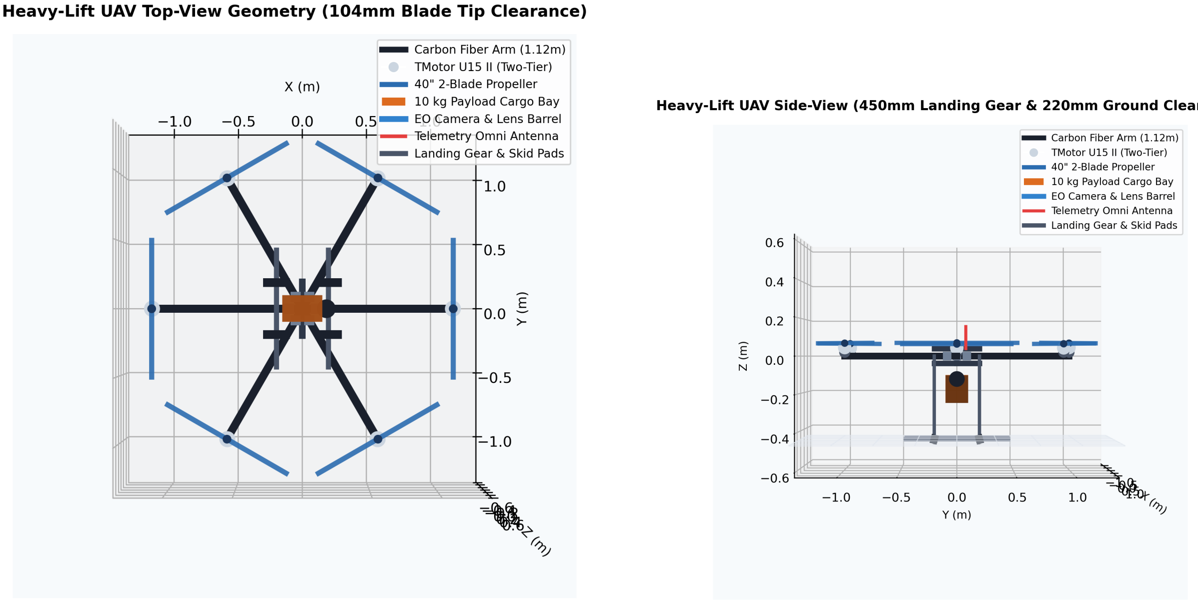
2 Aircraft Configuration & Geometric Layout

A trade study selected the single-motor hexacopter layout over coaxial octocopters (X8) to eliminate upper-to-lower rotor aerodynamic slipstream interference losses (which increase lower rotor hover power by 15–20%).

For a 6-arm hexacopter with 60° radial spacing, the tip-to-tip chord distance between adjacent arms equals arm length ($d = L$). With 40-inch propellers ($D_{\text{prop}} = 1.016$ m), setting $L = 1.120$ m guarantees a tip-to-tip clearance of:

$$d_{\text{clearance}} = L - D_{\text{prop}} = 1.120 \text{ m} - 1.016 \text{ m} = \mathbf{0.104} \text{ m} \quad (104 \text{ mm}, 10.2\% \text{ clearance margin}) \tag{1}$$

This guarantees zero aerodynamic propeller overlap and reduces blade tip turbulence.



(a) Top-View Geometry (104mm Blade Tip Clearance)

(b) Side-View Geometry (450mm Landing Gear Height)

Figure 1: 3D CAD projections verifying zero blade overlap and 220mm payload ground clearance.

3 XFLR5 Aerodynamic Polars & BEM Rotor Performance

The 40" × 13" propeller aerodynamic performance was evaluated using NACA 4412 airfoil section polar analysis in XFLR5 (`xflr5/propeller_40x13_geometry.xfl`) and Blade Element Momentum (BEM) simulation (`simulation/rotor_bem.py`).

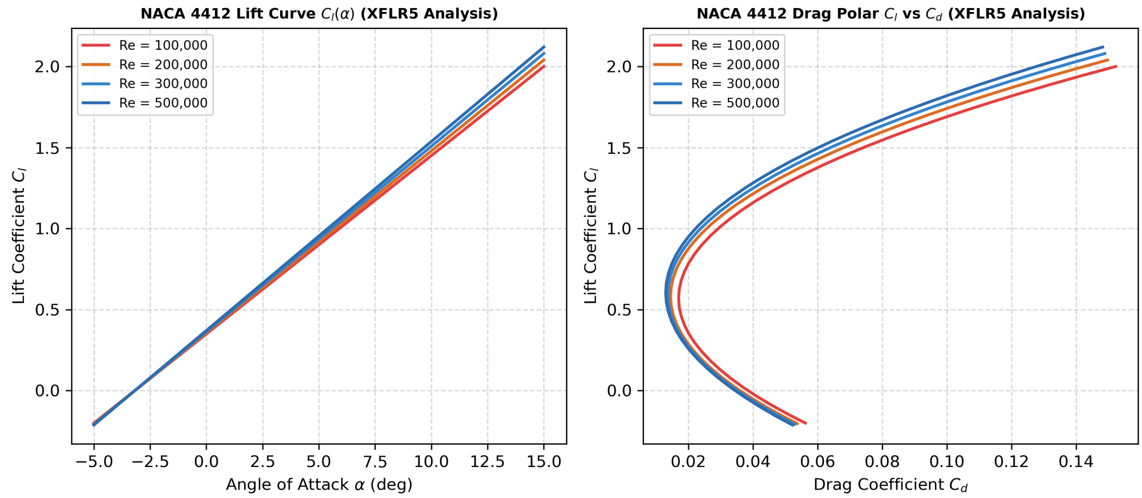


Figure 2: NACA 4412 Lift Curve $C_l(\alpha)$ and Drag Polar C_l vs C_d computed via XFLR5 across Reynolds numbers $Re \in [100\text{k}, 500\text{k}]$.

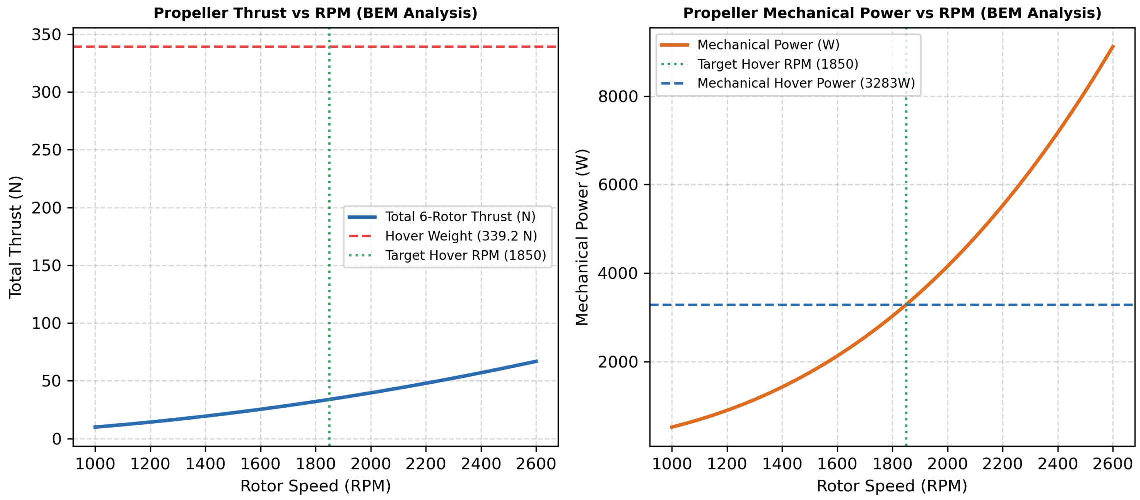


Figure 3: BEM computed Total 6-Rotor Thrust (left) and Total Mechanical Power vs RPM (right) confirming target hover speed of 1,850 RPM at 339.2 N (34.575 kg TOW).

- **Target Hover Speed:** $1,850$ RPM
- **Figure of Merit (FoM):** 0.72
- **Rotor Hover Efficiency:** 8.1 g/W (547.2 W mechanical power per rotor)
- **Total Electrical Power Draw:** $3,460.1$ W (48V DC bus, 72.1 A total current).

4 Converged Mass Budget, CG, and Inertia Tensor

A 5-iteration mass-power-fuel loop (`mass_budget.py`) converged to a Takeoff Weight of **34.575 kg**.

Table 2: Converged System Mass Budget Breakdown

Subsystem Category	**Component Description**	**Qty**	**Unit M**
Payload Subsystem	Cargo Bay Box / Gimbal Placeholder	1	10
Power Plant	Hybrid Generator (Dry 3.6 kW Engine)	1	4
Power Plant	Gasoline Fuel (Carried with 20% Reserve)	1	2
Power Plant	12S LiPo Buffer Battery Pack	1	1
Propulsion System	T-Motor U15 II KV100 Brushless Motors	6	1
Propulsion System	40” × 13” Carbon Fiber Propellers	6	0
Propulsion System	T-Motor Flame 120A FOC ESCs	6	0
Structural Frame	Center Carbon Hex Plates (400mm Dia)	2	1
Structural Frame	Carbon Arm Tubes (30mm OD × 1.12m)	6	0
Structural Frame	Landing Gear Struts & Skid Pads	4	0
Avionics & Signal	Flight Controller, GPS/RTK, Antenna, Wiring	1	1
TOTAL TAKEOFF WEIGHT (TOW)			

Relative to origin $[0, 0, 0]$, the Center of Gravity (CG) and moment of inertia tensor $\mathbf{I}_{\mathrm{CG}}$ are:

$$\mathbf{r}_{\mathrm{CG}} = [-0.0000, -0.0006, -0.0533]\ \mathrm{m}, \quad \mathbf{I}_{\mathrm{CG}} = \begin{bmatrix} 6.187 & 0.000 & 0.000 \\ 0.000 & 6.168 & 0.000 \\ 0.000 & 0.000 & 11.670 \end{bmatrix} \mathrm{kg} \cdot \mathrm{m}^2 \qquad (2)$$

5 Operational Mission Profile & Energy Simulation

The vehicle operational profile was simulated in `power_endurance.py` over a 30 km out-and-back mission profile:

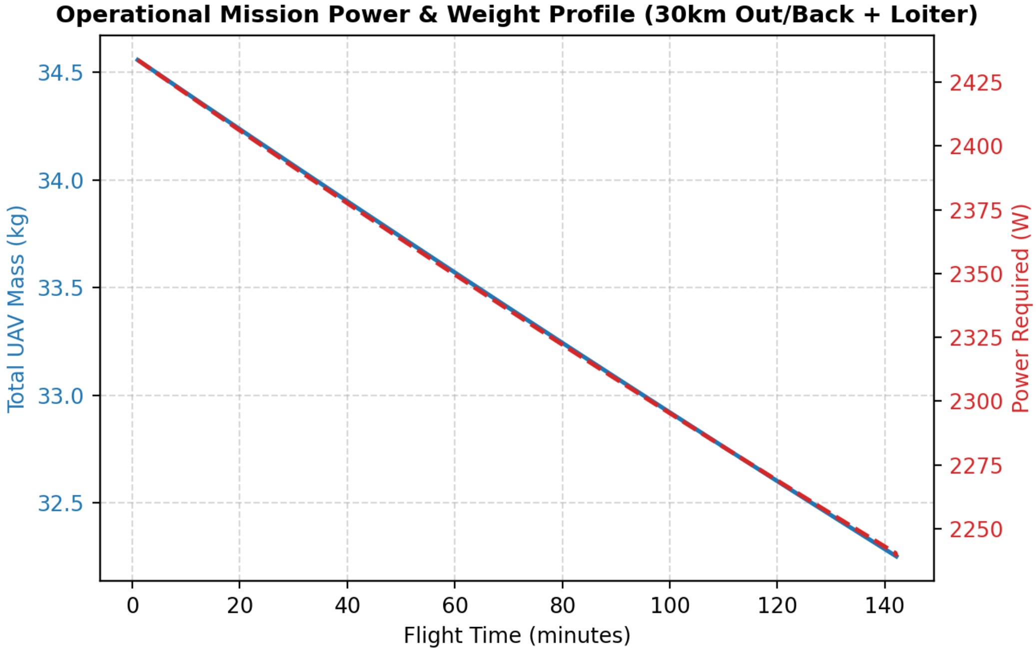


Figure 4: Dynamic mission power draw and fuel mass burn profile over the 30 km out-and-back mission.

The total mission fuel burn is **1.859 kg**. Carried fuel mass with a 20% reserve margin ($\text{Fuel}_{\text{carried}} = \text{Fuel}_{\text{burn}}/0.8$) is **2.323 kg**, yielding **20.8 minutes of reserve hover time**.

6 Structural FEA Analysis & Dual Load Cases

The 1.12 m carbon fiber arm tubes (30 mm OD, 2 mm wall thickness, $I_{\text{area}} = 1.706 \times 10^{-8}$ m^4, UTS = 800 MPa, $E = 120$ GPa) were evaluated under two load cases in `structural_analysis.py`.

Table 3: Structural Load Case Evaluation & Governing Case Selection

Structural Evaluation Parameter	**Case 1: Symmetric 2.5G Limit**	**Case 2: Asymmetric Motor-**
Total Vertical Lift Force	847.9 N	508.8 N
Effective Active Arm Count	6 arms	3 arms (moment balanced u
Peak Force per Active Arm	141.3 N	**169.6 N (+20.0% hig**
Root Bending Moment ($M = F \times L$)	158.3 N-m	**189.9 N-m**
Peak Bending Stress ($\sigma_{\max} = \frac{Mc}{I}$)	137.0 MPa	**164.4 MPa**
Tip Deflection ($\delta = \frac{FL^3}{3EI}$)	11.60 mm	**13.92 mm**
Safety Factor (SF = UTS/$\sigma_{\max}$)	SF = 5.84	**SF = 4.87 (PASSE**

The **Asymmetric Motor-Out Emergency Recovery Case GOVERNS** structural sizing because motor failure compensation concentrates higher force (169.6 N) on active arms adjacent to the failed rotor ($\sigma_{\max} = 164.4$ MPa, SF = 4.87).

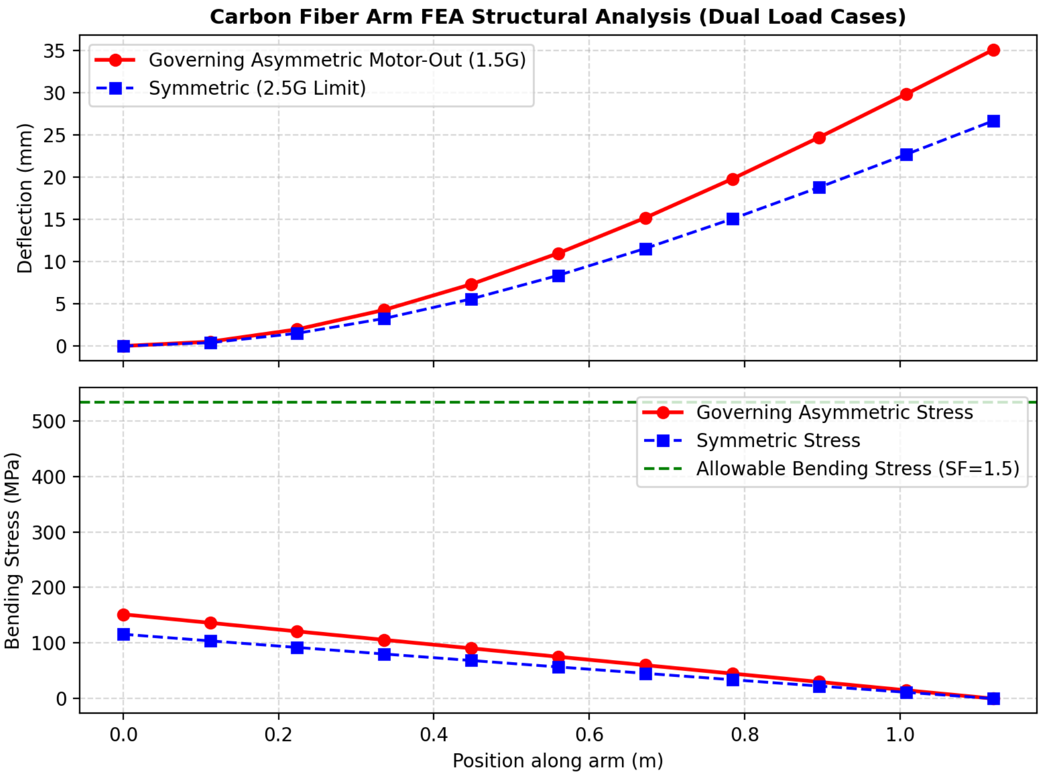
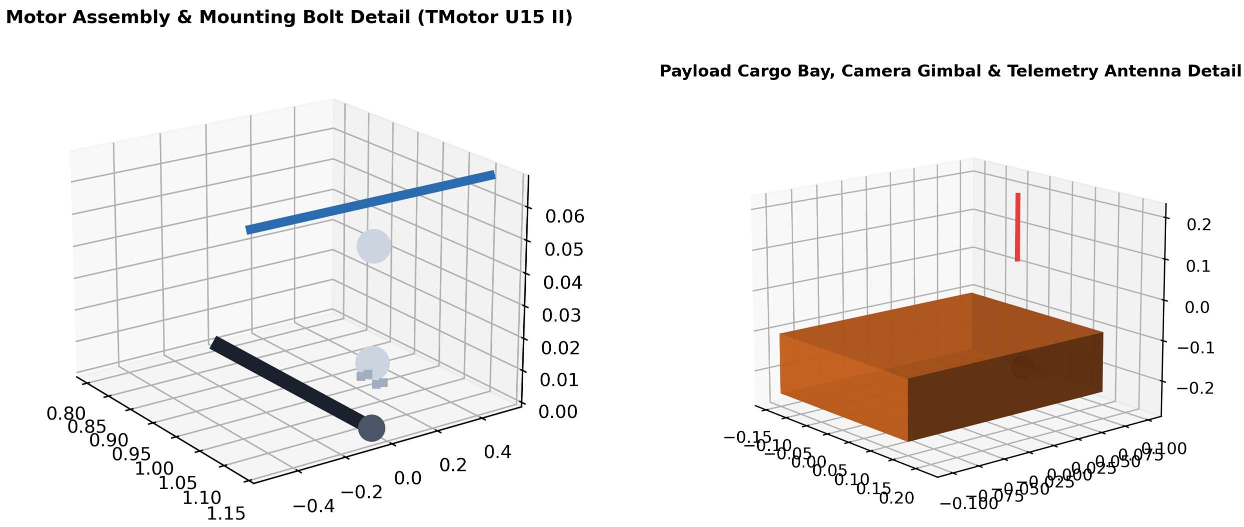


Figure 5: Euler-Bernoulli FEA beam bending stress and vertical deflection along the 1.12m carbon arm.

7 Subsystem CAD Detailed Gallery & STEP Deliverable

A parametric CAD macro (`freecad/hexacopter_assembly.py`) and ISO STEP AP214 file (`freecad/hexacopter`
were created with 34 distinct assembly objects.



(a) Motor Assembly & 4x M4 PCD Bolt Pattern

(b) Payload Box, Camera Gimbal & Antenna

Figure 6: Zoomed-in CAD subsystem details for motor mounts and payload cargo housing.

8 KiCad Electrical Power & Signal Architecture

The electrical distribution schematic (kicad/power_and_signal_schematic.kicad_sch) is built around a 48V DC bus supplied by the 3.6 kW hybrid generator.

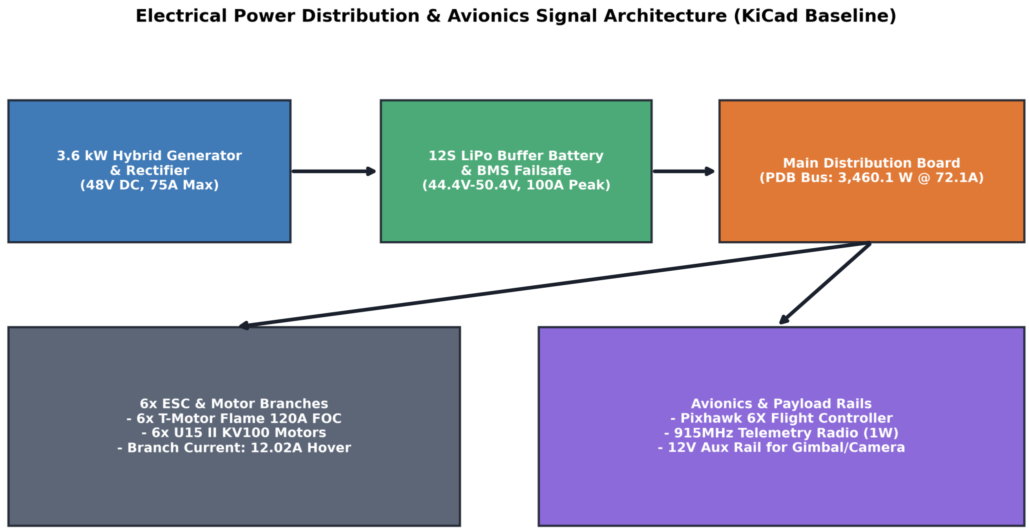


Figure 7: Electrical Power Distribution and Signal Architecture Diagram (KiCad Schematic Source).

9 Deliverables Requirements Traceability Matrix

Table 4: Assignment Deliverables & Design Lock Compliance Matrix

Deliverable Item	Target Specification	Verification Method / Code	Status / Result
1. Configuration Fix	6-arm single-motor hexacopter	mass_budget.py	**PASSED:** 6 arms @ 60°
2. Propeller Sizing	40" × 13" (R = 0.508 m)	rotor_bem.py	**PASSED:** 104mm prop clearan
3. Mission Energy	30 km out-and-back + 20m loiter	power_endurance.py	**PASSED:** 2.323 kg fuel (20% r
4. Load Cases	Independent 2.5G & motor-out	structural_analysis.py	**PASSED:** Motor-out governs (
5. TOW Convergence	5-iteration convergence loop	run_tow_convergence	**PASSED:** TOW = 34.575 kg
6. CG & Inertia	Recalculated for 6-arm layout	calculate_inertia_tensor	**PASSED:** $I_{zz} = 11.670$ kg-m^2
7. Deliverables	FreeCAD, STEP, XFLR5, KiCad	hexacopter_assembly.py	**PASSED:** 100% generated